ACRME — CRG & CR Foundation Provisioning Plan

Document Control

Field	Value
Title	ACRME — Capacity Reservation Group & Capacity Reservation Foundation Provisioning Plan
Version	1.0
Date	9 Sep 2026
Status	Draft — planning gate (pre-deployment)
Owner	Platform Engineering / Cloud Architecture
Source baseline	`acrme_requirements_baseline_v2_4.md` (authoritative: CAP-001..024, OPS-006, ENV-003, QUA-*, REG-003)
Related	`acrme_cr_crg_deployment_layout.md`, `acrme_poc_workbook_v2_4.md`, ADR-002/003/006
Purpose	Plan the creation of CRGs and CRs (the foundation) **before** any deployment — cover all grounds

> **Scope of this plan.** This is a **planning artifact for the creation phase only** — standing up the CRG/CR structure (groups + seed reservations at count 0), not deploying workloads and not building the reconciliation engine. It answers: what do we create, in what order, with what names, where, and how do we prove it is correct before we start deploying? It deliberately stops at the point where the seed structure exists and validates; workload deployment, reconciliation automation (CAP-005/CAP-006), and DR activation are downstream phases.

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1. Purpose & Position in the Programme

This plan operationalizes the **deployment layout** (`acrme_cr_crg_deployment_layout.md`) into an **ordered, checkable build sequence**. Where the layout doc answers "what does the structure look like", this plan answers "how do we bring it into existence correctly, and how do we know we covered everything."

Explicit goals:
1. Establish the canonical **order of operations** for creating CRGs and CRs (Azure-first, CAP-002).
2. Define the **seed matrix** to create (count-0 reservations per eligible SKU × region × AZ, CAP-022).
3. Apply the **naming convention** deterministically (OPS-006 / C-12).
4. Enforce **structural isolation** at creation time (per-AZ CRGs CAP-023; environment separation ENV-003).
5. Gate creation with **eligibility checks** (CAP-020/021 — no Availability-Set VMs; zonal placement).
6. Provide a **coverage matrix** proving every relevant requirement maps to a build step and a validation.
7. Define a **phased rollout** (pilot → geography-by-geography) so we do not big-bang the estate.

Non-goals (downstream phases, not this plan):
- Reconciliation automation / the container-app job (CAP-005/CAP-006) — Phase 3.
- Workload (VM/VMSS/AKS) deployment against reservations — Phase 2+.
- Quota pooling automation (QUA-003/004/008/009) — parallel track, prerequisite only here.
- DR activation runbook (DR-005/006/009) — separate DR programme.
- IaC authoring (Bicep/Terraform) — follows plan approval (see §12 Next Steps).

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2. Guiding Principles for the Build

These baseline rules constrain **how** we create resources and must hold at every step:

#	Principle	Requirement	Build implication
1	**Azure-first, then config**	CAP-002	Create the CRG + reservation in Azure and validate it **before** activating it in deployment config / scope file. Never reference a reservation in config that does not yet exist in Azure.
2	**Seed at zero, never delete**	CAP-009, CAP-022	Every eligible SKU/AZ is created as a reservation at **quantity 0**. Reaching 0 is normal; deletion is a governed decommission (CAP-010), never part of the build.
3	**Structural zone isolation**	CAP-011, CAP-023	Zone isolation is enforced by **separate per-AZ CRGs**, not just accounting. One regional (non-zonal) CRG + one CRG per AZ, per environment.
4	**Environment separation**	ENV-003	Prod ≠ non-prod, Prod ≠ DR (never share a CRG or subscription boundary). DR may co-locate/share with non-prod only where PLC-010 permits.
5	**Eligibility before onboarding**	CAP-020, CAP-021	Only zonal-placed (or regional-only-where-SKU-lacks-zonal) resources are eligible. Availability-Set VMs are excluded and flagged for deallocate/redeploy.
6		OPS-006, C-12	

#	Principle	Requirement	Build implication
	Deterministic naming		Every RG/CRG/subscription name is generated from tokens + zero-padded counter and validated against the convention; non-conforming names are governance exceptions.
7	**Budget-governed matrix**	CAP-022	A SKU/AZ enters the seed matrix only with a named owning product team + approved budget line. Anything scaled above 0 incurs cost.
8	**Quota is parallel, validated not re-gated**	QUA-001, QUA-007, ADR-006	Quota must be present in the deploying subscription (a build prerequisite we verify), but the build does not create quota gates — Azure enforces quota natively.

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3. Prerequisites — Pre-Flight Before Any Creation

Nothing is created until every item below is confirmed. (Aligns with the POC workbook Pre-Flight PF-01..PF-10.)

3.1 Identity, subscription & governance

ID	Prerequisite	How verified	Requirement
PRE-01	Target subscriptions exist and are named per convention (Prod / NonProd / DR per geography)	`az account list -o table`; name matches `sub-<org>-<domain>-<purpose>-<NN>`	OPS-006, ENV-003
PRE-02	Core subscription identi- fied and flagged **all-production**	Scope-file entry; CAP-001a classifica- tion set	CAP-001a
PRE-03	RBAC: build identity (UAMI/SPN) has least-privilege custom role at target RG/sub- scription scope	`az role assignment list --scope <SCOPE>`	GOV-*, security guide
PRE-04	Resource groups cre- ated per convention (`rg-odcr-<env>-<region>-<NN>`)	`az group show`; name validated	OPS-006
PRE-05	Region catalogue configuration loaded (distribution model, zone support, `DR_NOT_OFFERED` flags)	`PlacementPolicy` config version pinned	REG-001, REG-003
PRE-06	Scope-file schema ready (tenant, sub, region, AZ, RG, CRG, reservation name, SKU/family, env, buf- fer, enabled, effective date/version)	Scope-file present + versioned	CAP-001, CAP-019

3.2 Capacity, quota & preview features

ID	Prerequisite	How verified	Requirement
PRE-07	**Quota present** in each deploying subscription for the target VM families/regions	`az vm list-usage --location <REGION>` ≥ planned seed+buffer intent	QUA-002, QUA-007
PRE-08	Zone support confirmed per SKU × region (which SKUs support zonal reservations)	`az vm list-skus --location <REGION> --zone`	CAP-011, CAP-023
PRE-09	**Capacity Reservation Sharing** preview feature registration state recorded (if sharing in scope)	`az feature show` — record as Observed, not SLA	CAP-013, preview risk
PRE-10	**Quota Groups** preview state recorded (if pooling in scope)	`az feature show` — record as Observed	QUA-003, preview risk
PRE-11	No name-colliding CRGs already present	`az capacity reservation group list -g <RG>`	OPS-006
PRE-12	Eligible-SKU/AZ matrix drafted with owning **product team + budget line** per entry	Seed-matrix config reviewed & approved	CAP-022

> **Blocker rule.** If PRE-07 (quota) fails for a target family/region, do **not** create above-zero reservations there — seed at 0 is still valid, but flag a quota readiness risk (QUA-007) and raise a quota request (QUA-010) before any scale-up.

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4. Canonical Order of Operations (per environment × region)

This is the **repeatable creation sequence**. It is idempotent-friendly (re-running should not duplicate) and Azure-first (CAP-002).

```
FOR each (geography → environment → region) in the phased rollout:

  STEP 0  Pre-flight gate (Section 3) passes for this scope ............... GATE
  STEP 1  Set subscription context
            az account set --subscription <ENV_SUB>
  STEP 2  Ensure resource group (naming: rg-odcr-<env>-<region>-<NN>)
            az group create -n <RG> -l <REGION>            (if absent)
  STEP 3  Create the REGIONAL (non-zonal) CRG
            crg-<env>-<region>-reg
            az capacity reservation group create -g <RG> -n <CRG_REG> -l <REGION>
  STEP 4  Create one PER-AZ CRG for each supported zone
            crg-<env>-<region>-az1 / az2 / az3
            az capacity reservation group create -g <RG> -n <CRG_AZn> -l <REGION> --
zones n
  STEP 5  For each eligible SKU in the seed matrix (CAP-022):
            5a  Zonal SKU  → create count-0 reservation in each per-AZ CRG
                  az capacity reservation create -g <RG> \
                    --capacity-reservation-group <CRG_AZn> \
                    --name <CR_NAME> --sku <SKU> --capacity 0 \
                    --location <REGION> --zone n
            5b  Non-zonal SKU → create count-0 reservation in the regional CRG
                  az capacity reservation create -g <RG> \
                    --capacity-reservation-group <CRG_REG> \
                    --name <CR_NAME> --sku <SKU> --capacity 0 --location <REGION>
  STEP 6  VALIDATE in Azure (Section 8 gates) ........................... GATE
            provisioningState=Succeeded; zones correct; capacity=0
  STEP 7  ONLY AFTER Azure validation passes → activate in scope file
            (write CRG + reservation entries, enabled=true, effective date/version)
  STEP 8  Record evidence (resource IDs, request IDs, timestamps) into build log
```

Why this order: STEP 7 (config activation) strictly follows STEP 6 (Azure validation) to honor CAP-002 — the deployment pipeline reads the scope file to decide reservation association, so a config entry must never point at a non-existent Azure object.

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5. Naming Plan (OPS-006 / C-12)

All tokens resolved deterministically; counter is zero-padded (`C-12` width configurable).

Resource	Pattern	Token meaning	Example
Resource group	`rg-odcr-<env>-<region>-<NN>`	env ∈ {prod,cval,nonprod,dr}; region = short code; NN = instance counter	`rg-odcr-prod-eus2-01`
Regional CRG	`crg-<env>-<region>-reg`	scope `reg` = non-zonal SKUs	`crg-pr-eus2-reg`
Per-AZ CRG	`crg-<env>-<region>-az<n>`	scope `az1/az2/az3` = zone-bound	`crg-pr-eus2-az1`
Capacity Reservation	`res-<env>-<region>-<scope>-<SkuCompact>-<NNN>`	scope = reg/az1..; SkuCompact = SKU w/o punctuation	`res-pr-eus2-az1-StandardD16sv5-001`
Subscription	`sub-<org>-<domain>-<purpose>-<NN>`	purpose ∈	`sub-jda-cld-core-01`

Region short-code table (examples — driven by REG-001 config):

Region	Code	Region	Code
West US 3	wus3	Switzerland North	swn
Central US	cus	Sweden Central	sdc
Canada Central	cac	Australia East	aue
East US 2	eus2	Australia Southeast	ause
East Asia	ea	Southeast Asia	sea
Saudi Arabia Central	sac	UAE North	uan

> **Reconciliation note.** The older POC workbook (`acrme_poc_workbook_v2_4.md`) shows a per-SKU CRG example (`crg-prod-<region>-<sku>`) that **predates CAP-023**. This plan follows the **authoritative v2.4 per-zone CRG structure** (one CRG per zone holding multiple SKUs), consistent with the deployment-layout reference. The engine validates managed resources against OPS-006 and flags non-conforming names as governance exceptions.

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6. CRG Structure Plan (CAP-023)

Per **environment × region**, create exactly this set (assuming a 3-zone region):

```
<env> in <region>:
  ├── crg-<env>-<region>-reg   ← 1 regional CRG (non-zonal SKUs only)
  ├── crg-<env>-<region>-az1   ← 1 per-AZ CRG (zone 1)
  ├── crg-<env>-<region>-az2   ← 1 per-AZ CRG (zone 2)
  └── crg-<env>-<region>-az3   ← 1 per-AZ CRG (zone 3)
= 4 CRGs per environment per 3-zone region
```

- **Regional CRG** holds **only** SKUs that do not support zonal reservations (PRE-08 determines this per SKU).
- **Per-AZ CRGs** hold zone-bound reservations → structurally enforce CAP-011 (zone-1 capacity never counted elsewhere).
- **Environments are never mixed** in a CRG (ENV-003). Separate RGs and, where required, separate subscriptions per environment.
- Adjust CRG count for regions with fewer/more zones (e.g., a 2-zone region → reg + az1 + az2 = 3 CRGs).

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7. Seed Matrix Plan (CAP-022, CAP-020/021)

7.1 What gets seeded

Every **eligible** SKU/family × region × AZ combination in the approved scope file is created as a **count-0 reservation** ("seed reservation") in the correct CRG, so reconciliation can scale it from a known baseline the instant demand appears (CAP-007).

7.2 Eligibility gate (create only if ALL true)

Gate	Rule	Requirement
E1	SKU × region combination is in the approved scope file	CAP-001, CAP-019
E2	Placement is **zonal** (per-AZ CRG) OR SKU has no zonal support → regional CRG only	CAP-011, CAP-023
E3	Target is **not** an Availability Set (mutually exclusive with reservations)	CAP-020
E4	Any existing VM to onboard is in an eligible placement, else record deallocate/redeploy-to-AZ action first	CAP-021
E5	Entry has a named owning **product team** + approved **budget line**	CAP-022

7.3 Seed-matrix worksheet (template — fill per region before build)

CRG	Reservation Name	SKU	Zonal?	Initial Qty	Product Team	Budget Line	Eligibility (E1-E5)
crg-pr-eus2-az1	res-pr-eus2-az1-StandardD16s-v5-001	Standard_D16s_v5	Yes	0	Platform-Core	FY27-Q1-Compute-001	✅
crg-pr-eus2-az2	res-pr-eus2-az2-StandardD16s-v5-001	Standard_D16s_v5	Yes	0	Platform-Core	FY27-Q1-Compute-001	✅
crg-pr-eus2-az3	res-pr-eus2-az3-StandardD16s-v5-001	Standard_D16s_v5	Yes	0	Platform-Core	FY27-Q1-Compute-001	✅
crg-pr-eus2-reg	res-pr-eus2-reg-Standard-M128s-001	Standard_M128s	No (regional)	0	Data-Services	FY27-Q1-Compute-002	✅
…	…	…	…	0	…	…	…

7.4 Reactive discovery reconciled (CAP-024)

The build seeds the **known** matrix. When an allocated VM later appears for a SKU/AZ **not** yet seeded, the engine (downstream phase) auto-creates the CRG (if absent) + reservation at `allocated + buffer` **and** raises a scope-file governance item (CAP-019) to ratify it. This plan only needs to ensure the **naming + CRG structure rules are deterministic** so reactive creation slots in cleanly.

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8. Validation Gates (post-creation, before config activation)

Run these after STEP 5 and before STEP 7, per resource:

Gate	Check	Command / method	Pass criteria
V1	CRG provisioned	`az capacity reservation group show`	`provisioningState = Succeeded`
V2	Zones correct	same `show -o json`	per-AZ CRG `zones` = [n]; regional CRG has none
V3	Reservation provisioned	`az capacity reservation show`	`provisioningState = Succeeded`
V4	Seed quantity is 0	same `show`	`sku.capacity = 0`
V5	SKU matches	same `show`	`sku.name = <SKU>`
V6	Naming conforms	regex vs OPS-006 pattern	match; else governance exception
V7	Environment isolation	RG/CRG env token vs subscription purpose	no cross-env mix (ENV-003)
V8	Quota present for family/region	`az vm list-usage`	available ≥ planned scale-up intent (QUA-007)
V9	Scope-file consistency	scope file vs Azure inventory	1:1 after STEP 7 (CAP-019)
V10	Evidence captured	build log	resource IDs + request IDs + timestamps recorded

> A resource that fails any gate is **not** activated in the scope file (STEP 7 withheld) until remediated.

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9. Full Build Inventory by Geography (phased)

Grounded in the deployment-layout reference and the region catalogue (REG-001). CRG counts assume 3-zone regions (adjust per actual zone support, PRE-08).

Geography	Model	Environ-ments to build	Regions	CRGs (per env × 4)	Notes
US	3-region	Prod, CVAL, DR (distinct regions)	West US 3 / Central US / Canada Central	12 total	Only geography with full 3-way region separation
Europe	2-region	Prod; CVAL+DR co-located	Sweden Central (Prod) / Switzerland North (CVAL+DR)	12 total (4 Prod + 4 CVAL + 4 DR; CVAL & DR share region, **separate subs/CRGs**)	Co-location mandatory (PLC-010a); `cval_region == dr_region`
Australia	2-region	Prod; CVAL+DR co-located	Australia East (Prod) / Australia Southeast (CVAL+DR)	12 total	Same co-location pattern as EU
Asia Pacific	2-region	Prod; CVAL+DR co-located	East Asia (Prod) / Southeast Asia (CVAL+DR)	12 total	Japan East pending (REG-001) — not built until confirmed
Middle East	2-region	Prod; CVAL (optional)	Saudi Arabia Central (Prod) / UAE North (CVAL)	8 total (4 Prod + 4 CVAL); **no DR**	`DR_NOT_OFFERED` (DEC-001) — build **no DR CRGs**; seed record `dr_region = null`

> **Two-region reminder (memory-checked).** In EU/Australia/Asia Pacific, one region hosts **Prod** and the other hosts **CVAL + DR co-located** — DR **is** offered in-geo. `DR_NOT_OFFERED` is **Middle-East-specific** (legal/residency), not a general two-region outcome. Do not skip DR CRGs for EU/AU/APAC.

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10. Phased Rollout

Phase	Scope	Exit criteria
Phase 0 — Pre-flight	All PRE-01..PRE-12 for pilot scope	Every prerequisite green for the pilot region
Phase 1 — Pilot (single region, Prod)	One geography, Prod env only, one region, small eligible SKU set — e.g. US Prod West US 3	POC-01..POC-05 pass; V1..V10 pass; scope file consistent
Phase 2 — Pilot environments complete	Add CVAL + DR for the pilot geography (validates co-location for a 2-region geo, and 3-way separation for US)	Co-location accounting verified (HC-6/HC-7); ENV-003 isolation proven
Phase 3 — Geography rollout	Repeat Phases 1-2 per geography per the inventory (§9)	All in-scope geographies seeded + validated
Phase 4 — Handoff to reconciliation	Enable the reconciliation job to manage the seeded matrix (CAP-005/006)	Engine scales seeds from 0 on real demand; out of scope for this plan

> **Start small:** Phase 1 deliberately limits to one region + Prod + a handful of SKUs so the order-of-operations, naming, and validation gates are proven before any fan-out.

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11. Coverage Matrix — “Are We Covering Grounds?”

Every relevant creation-phase requirement mapped to a plan step and its validation. (Downstream-only requirements are marked as such.)

Requirement	Covered by	Validated by
CAP-001 authoritative scope file	§3 PRE-06, §4 STEP 7	V9
CAP-001a core = all production	§3 PRE-02	V7
CAP-002 Azure-first then config	§4 STEP 6→7 ordering	V9
CAP-009 zero-not-delete	§2 P2, §7.1 seed at 0	V4
CAP-011 zone isolation	§6 per-AZ CRGs	V2
CAP-013/014/015 sharing	Prereq PRE-09 (build seeds structure; sharing enabled downstream)	PRE-09 recorded
CAP-016 pre-deploy validation	(downstream — deploy phase)	n/a this plan
CAP-019 scope-file governance	§4 STEP 7, §7.4	V9
CAP-020 Availability-Set ineligible	§7.2 E3	E3 gate
CAP-021 deallocate/redeploy-to-AZ	§7.2 E4	E4 gate
CAP-022 seed matrix + budget	§7 whole section	E5 gate, V4
CAP-023 regional + per-AZ CRG	§6 structure	V1, V2
CAP-024 reactive discovery	§7.4 (deterministic naming enables it)	V6
ENV-003 environment separation	§6, §9 separate subs/RGs/CRGs	V7
OPS-006 / C-12 naming	§5 naming plan	V6
QUA-002/007 quota present	§3 PRE-07	V8
REG-001/003 catalogue + distribution	§3 PRE-05, §9 inventory	inventory review

Requirement	Covered by	Validated by
PLC-010a co-location (2-region)	§9 notes	Phase 2 co-location check
DEC-001 Middle East DR_NOT_OFFERED	§9 (no DR CRGs for ME)	inventory review
DR-003/004 lean DR (seed at 0/bootstrap)	§7.1 seed at 0	V4
HC-6/HC-7 co-location capacity floor	Phase 2 exit criteria	Phase 2 check

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12. Risks, Preview Dependencies & Out-of-Scope

12.1 Risks / dependencies

- **Preview features** (Capacity Reservation Sharing, Quota Groups) are **not** Microsoft contractual commitments — record observed behavior as Observed, not SLA (PRE-09/PRE-10). Sharing (Tier 2/3) build steps are gated on preview availability.
- **Quota absence** blocks above-zero scale-up (not seeding at 0). Raise quota requests (QUA-010) early for pilot families.
- **Zone support varies by SKU/region** (PRE-08) — a SKU without zonal reservation support must go in the regional CRG, not a per-AZ CRG; mis-placement fails V2.
- **Availability-Set legacy VMs** require deallocate/redeploy before onboarding (CAP-021) — service-impacting, owned by the workload team, not auto-migrated.

12.2 Explicitly out of scope for this plan

- Reconciliation automation / container-app job (CAP-005/006) — Phase 4.
- Workload deployment + reservation association (POC-03 onward is validation, not production deploy).
- Quota pooling/reclamation automation (QUA-003/004/008/009).
- DR declaration/activation runbook (DR-005/006/009).
- IaC authoring — see Next Steps.

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13. Next Steps (after this plan is approved)

1. **Fill the seed-matrix worksheet** (§7.3) for the Phase 1 pilot region with real SKUs, product teams, and budget lines.
2. **Confirm pilot scope** — recommended: **US Prod, West US 3**, 3-5 eligible SKUs.
3. **Author IaC** (Bicep or Terraform) implementing §4 order-of-operations + §5 naming, or a scripted `az` runbook mirroring the POC workbook Group 1.
4. **Execute Phase 0 pre-flight** and record evidence.
5. **Run Phase 1** and capture V1..V10 results before any fan-out.

> Tell me which of steps 1-3 to start on and I'll produce it (e.g., the Bicep/Terraform module, the seed-matrix worksheet pre-filled for a chosen pilot region, or a scripted `az` build runbook).

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14. Revision History

Version	Date	Change
1.0	9 Sep 2026	Initial CRG/CR foundation provisioning plan — order of operations, naming, CRG structure, seed matrix, eligibility gates, validation gates, phased rollout, coverage matrix. Reconciled to baseline v2.4.

End of document.